The number of cycles in a random permutation

Terence Tao · 23 November 2011

https://terrytao.wordpress.com/2011/11/23/the-number-of-cycles-in-a-random-permutation/

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Let n be a natural number, and let $\sigma : \{1,...,n\} \to \{1,...,n\}$ be a permutation of $\{1,...,n\}$, drawn uniformly at random. Using the cycle decomposition, one can view σ as the disjoint union of cycles of varying lengths (from 1 to n). For each $1 \leq k \leq n$, let C_k denote the number of cycles of σ of length k; thus the C_k are natural number-valued random variables with the constraint

$$\sum_{k=1}^{n} kC_k = n. \tag{1}$$

We let $C := \sum_{k=1}^{n} C_k$ be the number of cycles (of arbitrary length); this is another natural number-valued random variable, of size at most n.

I recently had need to understand the distribution of the random variables C_k and C. As it turns out this is an extremely classical subject, but as an exercise I worked out what I needed using a quite tedious computation involving generating functions that I will not reproduce here. But the resulting identities I got were so nice, that they strongly suggested the existence of elementary bijective (or "double counting") proofs, in which the identities are proven with a minimum of computation, by interpreting each side of the identity as the cardinality (or probability) of the same quantity (or event), viewed in two different ways. I then found these bijective proofs, which I found to be rather cute; again, these are all extremely classical (closely related, for instance, to Stirling numbers of the first kind), but I thought some readers might be interested in trying to find these proofs themselves as an exercise (and I also wanted a place to write the identities down so I could retrieve them later), so I have listed the identities I found below.

1. For any $1 \leq k \leq n$, one has $\mathbf{E}C_k = \frac{1}{k}$. In particular, $\mathbf{E}C = 1 + \frac{1}{2} + ... + \frac{1}{n} = \log n + O(1)$.
2. More generally, for any $1 \leq k \leq n$ and $j \geq 1$ with $jk \leq n$, one has $\mathbf{E}\binom{C_k}{j} = \frac{1}{k^j j!}$.
3. More generally still, for any $1 \leq k_1 < ... < k_r \leq n$ and $j_1, ..., j_r \geq 1$ with $\sum_{i=1}^{r} j_i k_i \leq n$, one has
$$\mathbf{E}\prod_{i=1}^{r}\binom{C_{k_i}}{j_i} = \prod_{i=1}^{r}\frac{1}{k_i^{j_i} j_{(i)!}}.$$
4. In particular, we have *Cauchy's formula*: if $\sum_{k=1}^{n} j_k k = n$, then the probability that $C_k = j_k$ for all $k = 1, ..., n$ is precisely $\prod_{k=1}^{n} \frac{1}{k^{j_k} j_{(k)!}}$. (This in particular leads to a reasonably tractable formula for the joint generating function of the C_k, which is what I initially used to compute everything that I needed, before finding the slicker bijective proofs.)
5. For fixed k, C_k converges in distribution as $n \to \infty$ to the Poisson distribution of intensity $\frac{1}{k}$.
6. More generally, for fixed $1 \leq k_1 < ... < k_r$, $C_{k_1}, ..., C_{k_r}$ converge in joint distribution to r independent Poisson distributions of intensity $\frac{1}{k_1}, ..., \frac{1}{k_r}$ respectively. (A more precise version of this claim can be found in this paper of Arratia and Tavaré.)
7. One has $\mathbf{E}2^C = n + 1$.
8. More generally, one has $\mathbf{E}m^C = \binom{n+m-1}{n}$ for all natural numbers m.

Comments (23)

 **Greg Martin** · 23 November 2011

Terry – are the distributions of the lengths of the longest cycle and shortest cycle also known?

>  **Terence Tao** · 23 November 2011
>
> This paper of Lloyd and Shepp seems to answer such questions fairly definitively, though the answers are somewhat messy.

 **Qiaochu Yuan** · 23 November 2011

For what it's worth, the joint generating function of the C_k can be computed using Polya's enumeration theorem, and it also has an elegant explanation in terms of combinatorial species (see my blog posts here and here for details).

 **Anonymous** · 23 November 2011

Typo in the first paragraph. 'brandom' should be 'random'. *[Corrected, thanks – T.]*

 **Ben Green** · 24 November 2011

There is a play (!) on this kind of thing by our friend Andrew Granville and his sister.
http://www.maa.org/mathtourist/mathtourist_01_06_10.html
I recall the accompanying music, which was in 29 time, making me feel rather queasy.

>  **Greg Martin** · 25 November 2011
>
> 29 time?
>
> >  **Ben Green** · 27 November 2011
> >
> > Yes, as in 29 beats to the bar. So far as I recall.

>  **Emmanuel Kowalski** · 26 November 2011
>
> The play was supposed to come out as a comics, but I don't know what happened to that project… More mathematically, the book of Arratia, Barbour and Tavaré, "Logarithmic combinatorial structures" (see http://www.ems-ph.org/books/book.php?proj_nr=15) contains many generalizations, treated in a uniform manner.
>
> >  **Jay Beder** · 12 August 2019
> >
> > The graphic novel has come out: Prime Suspects, by Andrew Granville and Jennifer Granville, with illustrations by Robert J. Lewis, Princeton U. Press. There are extensive endnotes by A Granville and others. Granville mentions the log N result, though without any explanation. (BTW, Tao, Green and Ziegler are among the characters in the novel.)

 **Lam. S.** · 28 November 2011

Flajolet and Knuth studied a similar problem in the graph theory context:
Click to access RR-0888.pdf

 **A central limit theorem for the determinant of a Wigner matrix « What's new** · 28 November 2011

[…] complicated (and uses facts about the distribution of cycles in random permutations, mentioned in this previous post), but one can compute that is comparable to for GUE and for GOE. (The discrepancy here comes […]



kate · 30 November 2011

Love your blog. Hardcore math I see.



Slipper.Mystery · 3 December 2011

One word puzzle that employs a property of random permutations, specifically the (relatively low) probability of large cycles, is the 100 prisoners. This wikipedia entry (not among those you've linked) contains other useful identities.
See also
condemned prisoners from someone who liked your reprise of the blue-eyed islanders.
The problem also shows up in this
tribute to Martin Gardner.



john mangual · 6 December 2011

I wonder what application you have in mind. There are lots of permutation group actions whose orbit structure we might be interested in.



Manjunath Krishnapur · 7 January 2012

The chinese restaurant construction of a uniform random permutation probably yields all these in the cleanest way – for example, the number of cycles has the same distribution as a sum of independent Bernoullis with parameters 1, 1/2, 1/ 3,...,1/n.



Vineet George · 26 December 2012

I have done extensive research on combination and permutation and found consistent and uniform result. This result which I have found is written on a book known as Junction (an art of counting combination and permutation). To view one of the result of my research work then log on to the site https://sites.google.com/site/junctionslpresentation/home
also see: https://sites.google.com/site/junctionslpresentation/proof-for-advance-permutation



The Poisson-Dirichlet process, and large prime factors of a random number | What's new · 21 September 2013

[...] Again, we prove this proposition below the fold. Now we turn to the second way (a topic, incidentally, that was briefly touched upon in this previous blog post): [...]



Sungjin Kim · 9 August 2014

While I prepare for TA session for probability class, I found that the first one $\mathbf{E}C_k = 1/k$ has a probabilistic proof:
Let A_i be the event that i is contained in a k-cycle. $1 \leq i \leq n$.
Then $kC_k = \sum 1_{A_i}$ where 1_A is the indicator function.
$\mathbf{E}kC_k = k\mathbf{E}C_k = \sum P(A_i) = \sum 1/n = 1$ follows from combinatorial counting argument.



Anonymous · 17 September 2018

Why is P(A_i) = 1/n?



Anonymous · 26 March 2019

For a permutation, you can rearrange its cycles, such that each cycle has its minimal at the end, and the minimal elements of the cycles are in increasing order.
For example, the permutation (125)(37) (in S_7) should be written as (251)(73)(4)(6).
Note that you can read that expression without the parentheses (i.e. the only way to get the order 2517346 is (251)(73)(4)(6)), in the following way: the first cycle should end at "1". the second cycle should end at the minimal element after the first cycle, and so on.

Hence, we have a bijection between permutations and orders of the numbers 1…n, so you can choose an uniform order.

But now, the length of the cycle of 1 is k, iff 1 is in the kth place in our order, and the probability for that is clearly 1/n, so P(A_1)=1/n . (and the same is true for any i)



Cycles of a random permutation, and irreducible factors of a random polynomial | What's new · 15 July 2015

[…] (Prime number theorem for permutations) A randomly selected permutation of will be a cycle with probability exactly . (This was noted in this previous blog post.) […]



Sungjin Kim · 28 June 2016

I think #8 holds in general for complex z as an application of Stirling number of first kind.



Counting objects up to isomorphism: groupoid cardinality | What's new · 13 April 2017

[…] a cardinality that converges in distribution to the Poisson distribution of rate (as discussed in this previous post), thus we see that the fixed points of a large random permutation asymptotically are distributed […]